Intergraph Smart 3D Hangers and Supports SmartParts Configuration

Hexagon Documentation

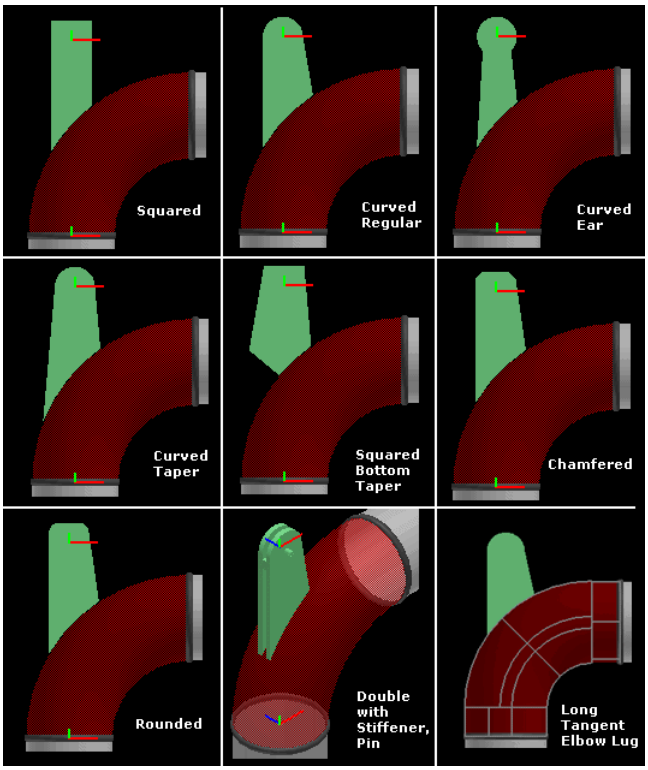
Generated 04/02/2026

Table of Contents

- Elbow Lug
  - Elbow Lug Properties

Elbow Lug

Library Name: SmartParts
Workbook: HS_S3DParts.xls
Codelist: HS_S3DParts_Codelist.xls
ProgID: HSSmartPart,Ingr.SP3D.Content.Support.Symbols.ElbowLug
Part Name: Elbow Lug
Part Description: Elbow Lug SmartPart



Prerequisites

Make sure that the following BulkLoad.xls workbooks in *[Product Folder]*\CatalogData\BulkLoad\DataFiles\ is already bulkloaded.

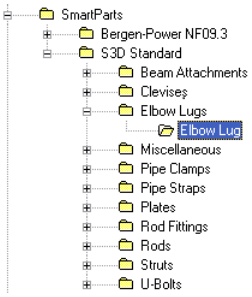
- AllCodeLists.xls
- AllCommon.xls
- GenericNamingRules.xls
- HS_System.xls
- HS_System_Codelist.xls
- HS_S3DParts_Codelists.xls

Sample SmartPart data is provided in HS_S3DParts.xls which is delivered in *[Product Folder]*\CatalogData\BulkLoad\DataFiles.

Part Placement

To place a sample Elbow Lug SmartPart delivered with the catalog:

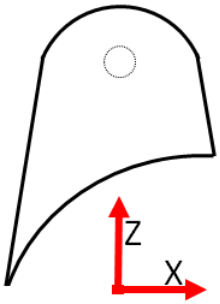
1. Click **Tasks > Hanger and Supports**.
2. Place a design support on a pipe.
3. Click **Place Part**, and then select the design support.
4. In the **Select Part** dialog box, select a part from **Parts > SmartParts > S3D Standard > Elbow Lugs > Elbow Lug** as shown below.



Part Properties

Millimeters (mm) is the default unit. Use "in" at the end of dimensional values to specify inches. Ports are shown as points highlighted in red.

Local Coordinate System



Elbow Lug Properties

IJUAhsRodDiameter::RodDiameter

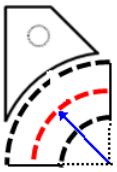
Specifies the rod diameter that is used by the elbow lug. This property does not affect the graphics and is used by the part selection rule.

IJOAhsPipeOD::PipeOD

Specifies the diameter of pipe that is used with the elbow lug.

IJOAhsElbow::ElbowRadius

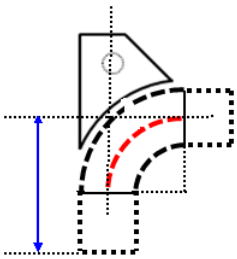
Specifies the radius of the elbow where the part is placed.



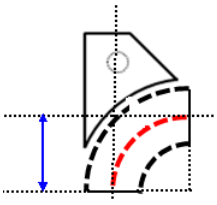
IJOAhsElbow::FaceToCenter

Specifies the distance between the face of the elbow and the center. For a short tangent elbow, face to center distance is same as the elbow radius.

Long Elbow

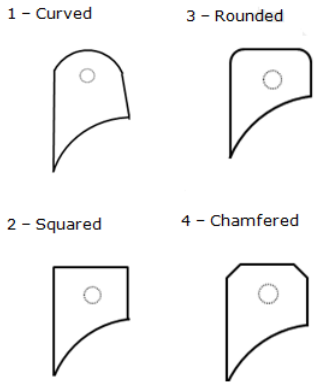


Short Elbow



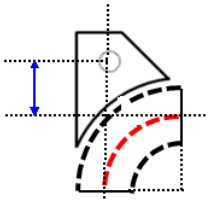
IJUAhsLug::TopShape

Specifies the graphic shape used for the top of the lug.



IJUAhsRodTakeOut::RodTakeOut

Specifies the distance between the horizontal pipe center line and the top lug port.

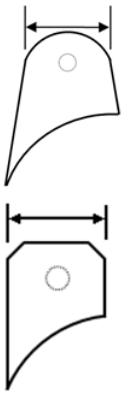


IJUAhsAngle1::Angle1

Specifies the angle between the top curve start and end point. If the top shape is not curved, then the software ignores this property. Angle1 must be positive, and greater than 90 degrees and less than 270 degrees. Else, a warning message displays and **Angle1** is reset to 180 degrees. If no value is assigned, a default value of 180 degrees is used.

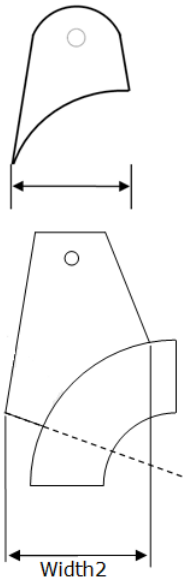
IJUAhsWidth1::Width1

Specifies the top width of the elbow lug. If **Width1** is set to zero, then, **Width2** is used. If **Width2** is also set to zero, then **PipeOD** is used.



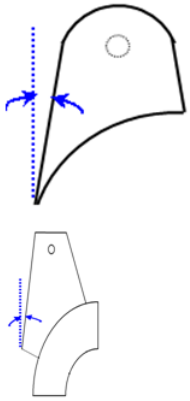
IJUAhsWidth2::Width2

Specifies the bottom width of the elbow lug. If this value is set to zero and **Angle 3** is set to zero, then **Width1** is used instead of **Width2**. If both the width values are set to zero, then **PipeOD** is used. If the left tapered portion does not intersect with the elbow, then the lug extends down along the pipe per **Angle4**. The bottom of the lug is drawn back to the pipe in line with the center of the radius of the turn with an angle of **Angle4.**



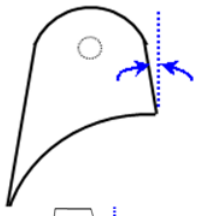
IJUAhsAngle2::Angle2

Specifies the angle of the left tapered portion. **Angle2** must be assigned a positive value and should less than 30 degrees. Else, an error message is displayed. If **Angle2** is set to zero, then the left line is drawn as global vertical.



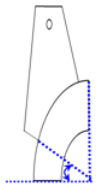
IJUAhsAngle3::Angle3

Specifies the angle of the right tapered portion. **Angle3** must be assigned a positive value less than 30 degrees. Else, an error message is displayed. **Angle3** overrides the **Wldth2** property to get the tapered bottom point.



IJUAhsAngle4::Angle4

Specifies the angle of the bottom tapered portion. **Angle4** must be assigned a positive value less than 45 degrees. Else, an error message is displayed. If the left tapered portion does not intersect with the elbow, then the lug extends down along the pipe per **Angle4.** If no value is assigned, **Angle4** is set to 45 degrees.

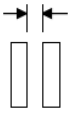


IJUAhsThickness1::Thickness1

Specifies the thickness of the lug.

IJUAhsGap1::Gap1

Specifies the gap between lug components of a double lug.



IJUAhsLug::StiffenerOffset

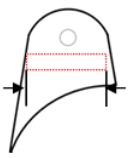
Specifies the offset of the stiffener between the two components of a double lug. **StiffenerOffset** is set from center of PIN port.

IJUAhsLug::StiffenerHeight

Specifies the height of the stiffener between the two components of a double lug.

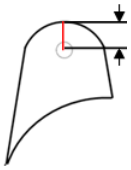
IJUAhsLug::StiffenerLength

Specifies the length of the stiffener between the two components of a double lug.



IJUAhsOffset1::Offset1

Specifies the distance between the top port and the top of the lug. When **TopShape** is set to 1 - Curved, **Offset1** is used for the eye radius.

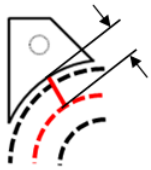


IJUAhsLug::ChamfLength

Specifies the length of the chamfered edges. This property is valid only when **TopShape** is set to 4 - Chamfered.

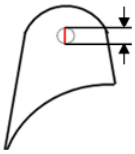
IJUAhsLug::RadiusActual

Specifies the required radius of the bottom arc of the elbow lug.



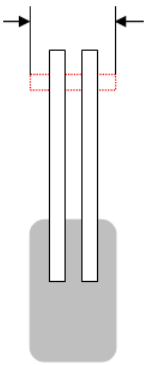
IJUAhsPin1::Pin1Diameter

Specifies the diameter of the pin. If this property is set to zero, then, pin is not drawn.



IJUAhsPin1::Pin1Length

Specifies the length of the pin. **Pin1Length** should be greater than **Gap1** + 2*__Thickness1__.



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